

Atharv Patil

Contact

patilatharv24@gmail.com
<https://github.com/patilatharv>
www.linkedin.com/in/atharvpatil30
(814) 600-6506

Education

The Pennsylvania State University - UP, Class of May 2025

BS in Computer Science with Minor in Math
College of Engineering

Technical Skills

Python, C++, C, TypeScript, Next.js, React.js, Java, JavaScript, SQL, Git, Node.js, Express.js, MongoDB, I2C, Verilog, HTML, CSS, ARKit, RealityKit, Swift, SwiftUI

Coursework

- Distributed Systems
- Database Management Systems
- Operating Systems
- Theory of Computation
- Object-Oriented Programming with Web-Based Applications
- Systems Programming
- Computer Organization and Design
- Data Structures and Algorithms
- Artificial Intelligence
- Programming Language Design

Clubs

HackPSU

Google Developer Student Club

Thon Hospitality CM

PSU Club Tennis

Work Experience

Amphenol, Nashua, NH
Software Engineering Intern

June - Aug 24

- Developed a **C++** CLI to interface with **I2C devices**, PWM and TC amplifier chips, leveraging their functionality for high-speed cable thermal testing
- Engineered robust **wrapper functions** for seamless access to all chip features per **datasheet specs**, enabling **reliable control and scalability**
- Created test setup with **PWM logic** to simulate real testing environment for **empirical data analysis** before integration into the **thermostream** system
- Programmed **customized PID control algorithm**, achieving precise temp correction from $\pm 7^{\circ}\text{C}$ to $\pm 0.5^{\circ}\text{C}$ to meet strict customer thresholds

Academic Excellence Center (AEC), Penn State

Oct 23 - Present

Lead Computer Science Tutor

- Provided academic support at AEC, by leading **exam reviews, one-on-one and group tutoring sessions** in **DSA, OOP, and systems programming**

Projects

NittanyBusiness, Full-Stack Development

Spring 25

- Designed an **end-to-end B2B e-commerce procurement platform** for SMEs with inefficient procurement process and limited supplier access
- Developed frontend and backend using **React.js, Node.js, and Express.js**, for key features like **user auth, product listings, and order management**
- Built and optimized **REST API** facilitating real-time data retrieval, reducing response times, ensuring efficient access to product and transaction data
- Designed **normalized logical database schema**, improving query efficiency and reduced redundant data storage

Peer-to-Peer File Sharing, Distributed Systems

Fall 24

- Designed a P2P file-sharing network with a central server and multiple peers in **Python**, enabling efficient, **decentralized file distribution**
- Developed a **chunk-based system** to split large files and allow **simultaneous downloads** from multiple peers, significantly improving download speed, with **multi-threading** to handle **parallel connections**
- Integrated file integrity checks using **hashing functions** to ensure the accuracy of downloaded chunks and discard corrupted data

Dynamic Storage Allocator, Memory Management

Spring 24

- Developed a dynamic memory allocator for **C programs** with custom implementations of **malloc, realloc, and free**, optimizing space utilization and throughput through **block header/footer optimization, free block coalescing, and size-based segregated linked lists**
- Created a **heap consistency checker** to track and resolve heap inconsistencies in debugging mode, ensuring reliable memory management

Linear Device Library, Systems Programming

Spring 23

- Designed and developed a Linux **mdadm utility** to configure 16 **JBOD** of 64 KB each into one coherent **linear device** of 1 MB using C
- Implemented **read and write functionality** for mdadm to access the storage architecture using a user-provided buf
- Optimized performance and increased flexibility by implementing **LRU cache** reducing JBOD operations cost by 30% and **networking support** by implementing client component of the embedded JBOD server over **TCP**